

A Production Learning That Made Sense Later: AG Failover (60s vs 5s)

About two years ago, I came across an interesting learning in my production environment. It wasn't exactly an issue, but rather a behavior I hadn't clearly understood at that time.

The Situation

Almost two years back, due to an unexpected network issue, an Availability Group (AG) failover occurred.

Immediately, we received an alert stating:

'Some users experienced a brief connection reconnect.'

We acknowledged the incident, explained that an AG failover had occurred, and informed stakeholders that we were working with the network team for RCA.

The Question from Users

Soon after, users raised an important question:

'Earlier, connections used to reconnect within 4–5 seconds during AG failover. Why did it take 40–60 seconds this time?'

At that point, based on my experience:

- During patching
- During planned or unplanned failovers

I had always observed 40–60 seconds for application connections to re-establish. So, the expectation of 4–5 seconds was surprising to me.

Deep Dive & Internal Discussion

Since this happened during my initial days on the project, I immediately connected with a senior team member and discussed the behavior in detail.

That's when an important point came up.

He explained that:

- The AG listener had multiple IPs
- Each IP belonged to a different subnet
- This was a true multi-subnet Availability Group

Which means:

Applications should reconnect in 2–5 seconds, not 40–60 seconds.

Root Cause Identified

After validating the configuration, we found that:

- Listener IPs were indeed in different subnets
- However, the application connection strings were missing:
- `MultiSubnetFailover=True`

Because of this:

- Clients attempted listener IPs sequentially
- Waited for TCP timeout + DNS delay
- Resulted in 40–60 seconds reconnection time

The Fix

We coordinated with the application team and updated the connection strings to include:

`MultiSubnetFailover=True`

The Result (Observed in Subsequent Failovers)

After the change:

- Failover reconnection time dropped to 2–5 seconds
- No DNS wait
- No TCP timeout delay
- Parallel connection attempts worked as designed

This behavior aligned perfectly with multi-subnet AG architecture.

Key Takeaway

`MultiSubnetFailover=True` is not optional for multi-subnet AGs, it is mandatory when listener IPs span different subnets.

Even with correct AG and cluster configuration:

- A missing connection string parameter can completely change the failover experience.
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How to Confirm a True Multi-Subnet AG

```
SELECT listener_id,  
       ip_address,  
       network_subnet_ip,  
       network_subnet_prefix_length  
FROM sys.availability_group_listener_ip_addresses;
```

Multi-subnet example:

- 10.10.10.x /28
- 189.16.60.x /28

If all listener IPs are from the same subnet, MultiSubnetFailover=True will not help
(And in 95% of environments, AGs are still single-subnet.)

Connection String Validation

```
Server=AGListenerName;  
Database=testdb;  
MultiSubnetFailover=True;  
ApplicationIntent=ReadWrite;
```